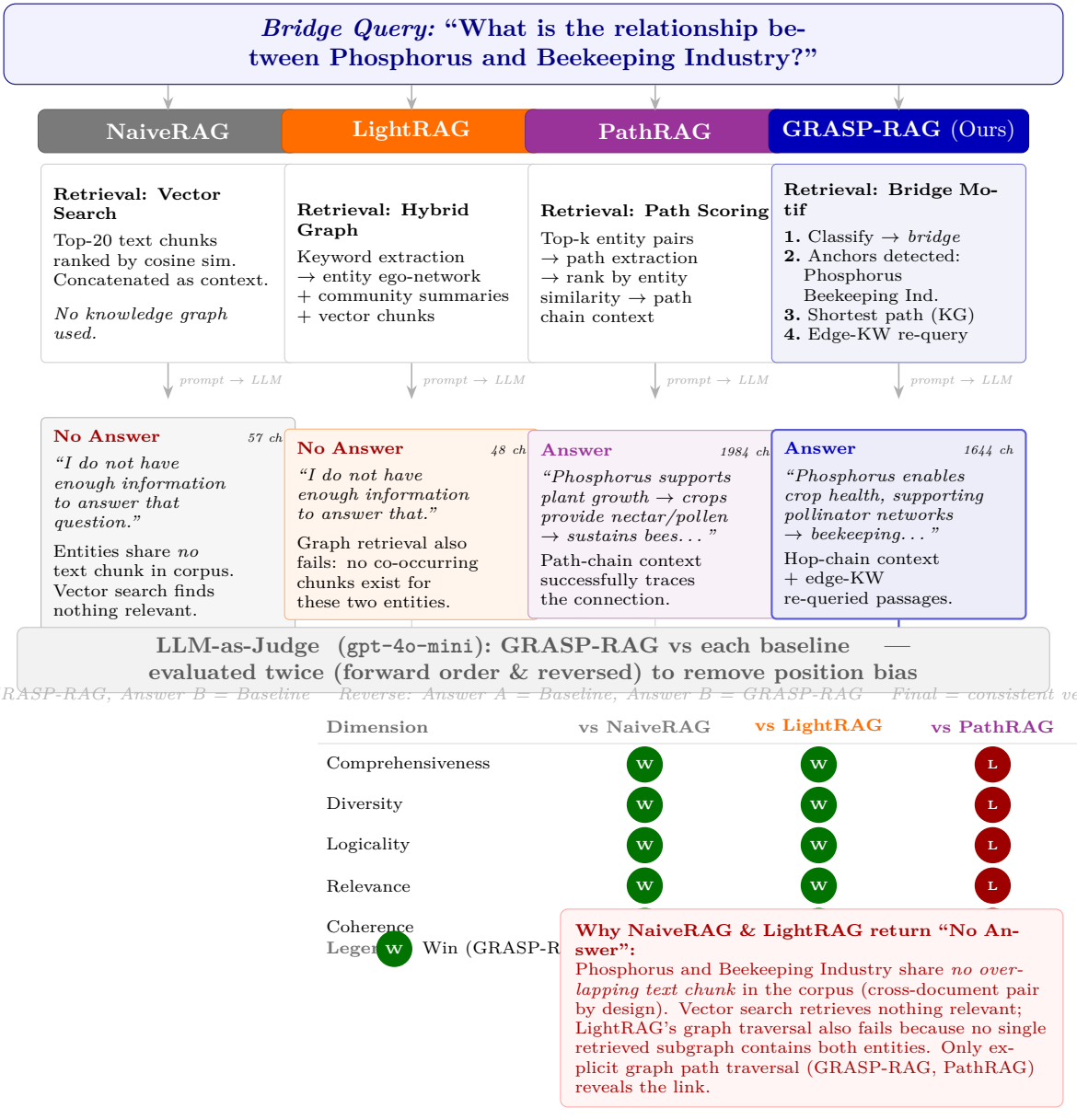

Bridge Query: "What is the relationship between Phosphorus and Beekeeping Industry?"
NaiveRAG
LightRAG
PathRAG
GRASP-RAG (Ours)
Retrieval: Vector Search
Top-20 text chunks ranked by cosine sim. Concatenated as context.
No knowledge graph used.
Retrieval: Hybrid Graph
Keyword extraction → entity ego-network + community summaries + vector chunks
Retrieval: Path Scoring
Top-k entity pairs → path extraction → rank by entity similarity → path chain context
Retrieval: Bridge Motif
1. Classify → *bridge*
2. Anchors detected: Phosphorus Beekeeping Ind.
3. Shortest path (KG)
4. Edge-KW re-query
prompt → LLM
prompt → LLM
prompt → LLM
prompt → LLM
No Answer 57 ch
"I do not have enough information to answer that question."
Entities share *no* text chunk in corpus. Vector search finds nothing relevant.
No Answer 48 ch
"I do not have enough information to answer that."
Graph retrieval also fails: no co-occurring chunks exist for these two entities.
Answer 1984 ch
"Phosphorus supports plant growth → crops provide nectar/pollen → sustains bees. . . "
Path-chain context successfully traces the connection.
Answer 1644 ch
"Phosphorus enables crop health, supporting pollinator networks → beekeeping. . . "
Hop-chain context + edge-KW re-queried passages.
LLM-as-Judge (gpt-4o-mini): GRASP-RAG vs each baseline — evaluated twice (forward order & reversed) to remove position bias
Forward: Answer A = GRASP-RAG, Answer B = Baseline Reverse: Answer A = Baseline, Answer B = GRASP-RAG Final = consistent verdict across both passes
Dimension
vs NaiveRAG
vs LightRAG
vs PathRAG
Comprehensiveness W W L
Diversity W W L
Logicality W W L
Relevance W W L
Coherence
Legend W Win (GRASP-R
Why NaiveRAG & LightRAG return "No Answer":
Phosphorus and Beekeeping Industry share *no overlapping text chunk* in the corpus (cross-document pair by design). Vector search retrieves nothing relevant; LightRAG's graph traversal also fails because no single retrieved subgraph contains both entities. Only explicit graph path traversal (GRASP-RAG, PathRAG) reveals the link.